Guided: Scripting for Remote SSH Class activity

Skills you will learn

Version Control & Linux: System Administration

Napkin - 3.22.240.226

Before Beginning - EOF

Sometimes, we have to execute or send commands to a remote server to configure it. For example, download something or update something. But sometimes, sending one line of code is just not enough.

In Linux, EOF stands for "End Of File." In the context of shell scripting, EOF is commonly used as a marker to indicate the start and end of a "here document" (heredoc). A heredoc allows you to pass multiple lines of input to a command. For example:

```
cat <<EOF
This is the first line.
This is the second line.
And this is the third line.
EOF
```

Copy Explain

> *Usually, EOF is used but you could use whatever name, like DIOGO or SECONDEOF.*

In this snippet, everything between the first `EOF` (after `<<`) and the ending `EOF` is provided as input to the `cat` command, which then outputs it to the terminal.

Example 2: Using a Heredoc in a Script

```
#!/bin/bash
echo "Starting the script..."
cat <<EOF
This is a message from the heredoc.
It can span multiple lines.
EOF
echo "Script completed."
```

Copy Explain

Save this as `myscript.sh`, make it executable (`chmod +x myscript.sh`), and run it. The heredoc block is treated as a single string input to the `cat` command.

> *This also means that we're able to send commands to be executed remotely.*

Actual LAB

This step-by-step lab guides students through creating and running local scripts that use SSH to execute remote commands on a VPS. The VPS in this example is accessible via:

```
ssh -i "ironhack-key.pem" ubuntu@3.12.148.48
```

Copy Explain

You can adapt it to your own server/keys.

> *This lab will help you grasp the concepts of **local** and **remote**. This is very important, as one day you'll be working in a network of 10, 100, 1000 virtual machines (called VPS normally, EC2 instances in AWS, or VM in Azure). So you need to be very aware of where you're working.*

Overview

1. **Local vs Remote Execution**
   - **Local script**: A script that resides and is run on your local machine (WSL, macOS, Linux, etc.).
   - **Remote commands**: Commands sent to and executed on the VPS via `ssh`.
   - **Who am I?** Use the following commands to figure out details about your remote session:

     ```
     whoami  # Know the logged-in user
     cat /etc/os-release  # Identify the OS
     ```

     Copy Explain

2. **SSH Key**
   - Assume you already have a key pair named `ironhack-key.pem` with the correct permissions:

     ```
     chmod 400 ironhack-key.pem
     ```

     Copy Explain

   - The VPS is configured to accept this key.
3. **Goals**
   - Practice running simple file-creation commands on the VPS via SSH from a local script.
   - Understand the usage of loops in shell scripts (specifically, a `for` loop).
   - Learn how to accept input (parameters) in shell scripts.
   - Work with environment variables in a script to define paths or other parameters.

Step 1: Create a Single File with Your Name

Objective

Run a local script that, through SSH, creates a single file on the VPS with your name, e.g., `diogo.txt`.

This script is to be run in your **local machine**. It creates a variable `MY_NAME`, then the script, on its own, connects to a remote server and executes a multi-line command inside the **remote machine**.

Example Script: `create_one_file.sh`

```
#!/bin/bash
# Usage: ./create_one_file.sh
MY_NAME="diogo"

ssh -i "ironhack-key.pem" ubuntu@3.143.5.172 << EOF
echo "Creating a file named ${MY_NAME}.txt on the VPS..."
touch ${MY_NAME}.txt
echo "File created in home directory of ubuntu user."
EOF
```

Copy Explain

How to Run

1. Give the script execute permissions:

```
chmod +x create_one_file.sh
```

Copy Explain

2. Run it:

```
./create_one_file.sh
```

Copy Explain

3. Check on the VPS that the file was created:

```
ssh -i "ironhack-key.pem" ubuntu@3.12.148.48
```

Copy Explain

IRONHACK

Step 2: Create Multiple Files Using a For Loop

Objective

Use a script that creates multiple files in the VPS, e.g., `diogo-1`, `diogo-2`, `diogo-3`, etc.

Example Script: `create_multiple_files.sh`

```
#!/bin/bash
```

Copy Explain

```
# Usage: ./create_multiple_files.sh
MY_NAME="diogo"

ssh -i "ironhack-key.pem" ubuntu@3.12.148.48 << EOF
echo "Creating multiple files on the VPS..."
for i in {1..5}
do
touch "${MY_NAME}-\$i"
echo "Created file: ${MY_NAME}-\$i"
done
echo "Done creating files."
EOF
```

How to Run

```
chmod +x create_multiple_files.sh
./create_multiple_files.sh
```

Copy Explain

Check the VPS:

```
ssh -i "ironhack-key.pem" ubuntu@3.12.148.48
ls diogo-*
exit
```

Copy Explain

Step 3: Accept a Name as Input and Create Multiple Files

Example Script: `create_files_with_name.sh`

```
#!/bin/bash
# Usage: ./create_files_with_name.sh <NAME>
NAME="$1"
if [ -z "$NAME" ]; then
  NAME="student"
fi
ssh -i "ironhack-key.pem" ubuntu@3.12.148.48 << EOF
echo "Creating files with name: $NAME"
for i in {1..3}
do
touch "$NAME-\$i"
echo "Created file: $NAME-\$i"
done
EOF
```

Copy Explain

How to Run

```
chmod +x create_files_with_name.sh
./create_files_with_name.sh test
```

Copy Explain

Check the VPS:

```
ssh -i "ironhack-key.pem" ubuntu@3.12.148.48
ls test-*
exit
```

Copy Explain

Challenge 3: Create an Interactive Script

Turn all the steps above into an interactive menu where you can select what you want to do, how many files, the name, even the path!

```
#!/bin/bash
echo "Please enter your name:"
read name
echo "Hello, $name!"
```

Copy Explain

AI TUTOR ACTIONS

Generate a practice quiz Summarize this unit Ask about this unit

Complete unit